Western Cape Soil Treatment & Watering Dosage Reference Guide

Official agricultural extension rules for soil pH amendment application rates (calcitic lime, dolomitic lime, and elemental sulfur) and weekly temperature/rainfall-based watering adjustments across sandy, sandy_loam, loam, clay_loam, and clay soils.

Table 1: Soil pH Amendment Dosages by Soil Type and Target Shift

rule_id	soil_type	current_ph_bracket	target_ph_shift	amendment_type	dosage_g_per_m2	application_notes
RULE-PH-01	sandy	5.0 - 5.5	+1.0 (to 6.0-6.5)	Agricultural Lime	100	Incorporate 100g/m2 calcitic lime into top 15cm of sandy soil 2 weeks before planting
RULE-PH-02	sandy_loam	5.0 - 5.5	+0.8 to +1.0 (to 6.0-6.8)	Agricultural Lime	150	Incorporate 150g/m2 calcitic lime into top 20cm of sandy_loam soil 2 weeks before planting
RULE-PH-03	loam	5.0 - 5.5	+0.8 to +1.0 (to 6.0-6.8)	Agricultural Lime	200	Incorporate 200g/m2 calcitic lime into top 20cm of loam soil 3 weeks before planting
RULE-PH-04	clay_loam	5.0 - 5.5	+0.8 to +1.0 (to 6.0-6.8)	Agricultural Lime	260	Incorporate 260g/m2 calcitic lime into top 20cm of clay_loam soil 3 weeks before planting
RULE-PH-05	clay	5.0 - 5.5	+0.8 to +1.0 (to 6.0-6.8)	Agricultural Lime	320	Incorporate 320g/m2 calcitic lime into top 20cm of heavy clay soil 4 weeks before planting
RULE-PH-06	sandy	5.5 - 6.0	+0.5 to +0.8 (to 6.3-6.8)	Agricultural Lime	70	Apply 70g/m2 calcitic lime and rake into top 10cm of sandy soil
RULE-PH-07	sandy_loam	5.5 - 6.0	+0.5 to +0.8 (to 6.3-6.8)	Agricultural Lime	100	Apply 100g/m2 calcitic lime into top 15cm of sandy_loam soil
RULE-PH-08	loam	5.5 - 6.0	+0.5 to +0.8 (to 6.3-6.8)	Agricultural Lime	140	Apply 140g/m2 calcitic lime into top 15cm of loam soil
RULE-PH-09	clay_loam	5.5 - 6.0	+0.5 to +0.8 (to 6.3-6.8)	Agricultural Lime	180	Apply 180g/m2 calcitic lime into top 15cm of clay_loam soil
RULE-PH-10	clay	5.5 - 6.0	+0.5 to +0.8 (to 6.3-6.8)	Agricultural Lime	220	Apply 220g/m2 calcitic lime into top 15cm of clay soil
RULE-PH-11	sandy	4.5 - 5.0	+1.5 (to 6.0-6.5)	Dolomitic Lime	160	Apply 160g/m2 dolomitic lime to supply magnesium and raise pH in acidic sand
RULE-PH-12	sandy_loam	4.5 - 5.0	+1.5 (to 6.0-6.5)	Dolomitic Lime	220	Apply 220g/m2 dolomitic lime into top 20cm of sandy_loam soil
RULE-PH-13	loam	4.5 - 5.0	+1.5 (to 6.0-6.5)	Dolomitic Lime	280	Apply 280g/m2 dolomitic lime into top 20cm of loam soil
RULE-PH-14	clay_loam	4.5 - 5.0	+1.5 (to 6.0-6.5)	Dolomitic Lime	360	Apply 360g/m2 dolomitic lime into top 20cm of clay_loam soil
RULE-PH-15	clay	4.5 - 5.0	+1.5 (to 6.0-6.5)	Dolomitic Lime	440	Apply 440g/m2 dolomitic lime into top 20cm of clay soil
RULE-PH-16	sandy	7.2 - 7.8	-0.8 to -1.0 (to 6.2-6.8)	Elemental Sulfur	40	Incorporate 40g/m2 elemental sulfur into top 15cm of sandy soil 4 weeks before planting
RULE-PH-17	sandy_loam	7.2 - 7.8	-0.8 to -1.0 (to 6.2-6.8)	Elemental Sulfur	60	Incorporate 60g/m2 elemental sulfur into top 15cm of sandy_loam soil
RULE-PH-18	loam	7.2 - 7.8	-0.8 to -1.0 (to 6.2-6.8)	Elemental Sulfur	90	Incorporate 90g/m2 elemental sulfur into top 15cm of loam soil
RULE-PH-19	clay_loam	7.2 - 7.8	-0.8 to -1.0 (to 6.2-6.8)	Elemental Sulfur	130	Incorporate 130g/m2 elemental sulfur into top 15cm of clay_loam soil
RULE-PH-20	clay	7.2 - 7.8	-0.8 to -1.0 (to 6.2-6.8)	Elemental Sulfur	170	Incorporate 170g/m2 elemental sulfur into top 15cm of clay soil
RULE-PH-21	sandy	7.8 - 8.5	-1.2 to -1.5 (to 6.5-7.0)	Elemental Sulfur	70	Incorporate 70g/m2 elemental sulfur and acidic compost into sandy soil
RULE-PH-22	sandy_loam	7.8 - 8.5	-1.2 to -1.5 (to 6.5-7.0)	Elemental Sulfur	100	Incorporate 100g/m2 elemental sulfur and pine needle mulch into sandy_loam soil
RULE-PH-23	loam	7.8 - 8.5	-1.2 to -1.5 (to 6.5-7.0)	Elemental Sulfur	150	Incorporate 150g/m2 elemental sulfur into loam soil
RULE-PH-24	clay_loam	7.8 - 8.5	-1.2 to -1.5 (to 6.5-7.0)	Elemental Sulfur	210	Incorporate 210g/m2 elemental sulfur into clay_loam soil
RULE-PH-25	clay	7.8 - 8.5	-1.2 to -1.5 (to 6.5-7.0)	Elemental Sulfur	270	Incorporate 270g/m2 elemental sulfur into clay soil

Table 2: Weekly Watering & Mulching Adjustment Rules by Climate Bracket

Weekly baseline watering adjustments (mm/week) and mulching protocols calculated against regional ambient temperature and rainfall.

rule_id	temp_bracket_c	rainfall_bracket_mm	soil_type	weekly_water_adjustment_mm	mulching_recommendation
RULE-WT-01	< 20	> 40	sandy	-15	Light bark mulch 2cm to prevent crusting
RULE-WT-02	< 20	> 40	sandy_loam	-20	Light compost mulch 2cm to maintain aeration
RULE-WT-03	< 20	> 40	loam	-20	No mulch required; keep bed aerated
RULE-WT-04	< 20	> 40	clay_loam	-25	Open drainage furrows; no mulch during deluge
RULE-WT-05	< 20	> 40	clay	-25	Avoid waterlogging; ensure ridge drainage
RULE-WT-06	< 20	10 - 40	sandy	0	Straw mulch 3cm to retain base moisture
RULE-WT-07	< 20	10 - 40	sandy_loam	0	Standard straw mulch 3cm
RULE-WT-08	< 20	10 - 40	loam	0	Organic compost layer 3cm
RULE-WT-09	< 20	10 - 40	clay_loam	-5	Light compost topdress
RULE-WT-10	< 20	10 - 40	clay	-5	Monitor rootzone saturation before watering
RULE-WT-11	20 - 28	< 10	sandy	+25	Heavy pine-bark mulch 7cm to stop evaporation
RULE-WT-12	20 - 28	< 10	sandy_loam	+20	Sugarcane or straw mulch 5cm
RULE-WT-13	20 - 28	< 10	loam	+15	Well-rotted compost mulch 5cm
RULE-WT-14	20 - 28	< 10	clay_loam	+10	Straw mulch 5cm to prevent surface cracking
RULE-WT-15	20 - 28	< 10	clay	+10	Shredded leaf mulch 5cm to keep soil workable
RULE-WT-16	> 28	< 10	sandy	+35	Deep mulch 8cm with drip irrigation twice daily
RULE-WT-17	> 28	< 10	sandy_loam	+30	Deep straw or woodchip mulch 7cm with shade cloth
RULE-WT-18	> 28	< 10	loam	+25	Dense organic mulch 7cm; water at dawn
RULE-WT-19	> 28	< 10	clay_loam	+20	Straw mulch 6cm to preserve moisture and stop baking
RULE-WT-20	> 28	< 10	clay	+15	Mulch 6cm; slow deep soaks to prevent runoff